Sarat Chandra Nagavarapu

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Professional Summary

A Senior Research Scientist with nearly ten years of experience in artificial intelligence, machine learning, optimization, and data analytics across intelligent transportation, aviation, robotics, and satellite systems. Experienced in developing predictive models, decision-support algorithms, and data pipelines for complex, safety-critical environments. Proven success leading R&D collaborations under the previous employers with Singapore government agencies, airlines, and academic partners, delivering operational prototypes and patented solutions. Strong background in problem-solving, predictive modelling, and large-scale data analysis. Committed to translating research prototypes into scalable AI systems and working with stakeholders to deliver data-driven solutions that improve efficiency and decision-making.

Employment History

2026-present **Senior Scientist-I**, *Advanced Remanufacturing and Technology Centre (ARTC), Agency for Science, Technology And Research (A*STAR)*, Singapore.

2023-2026 **Senior Scientist-I**, *Institute for Infocomm Research (I2R), Agency for Science, Technology And Research (A*STAR)*, Singapore.

2022-2023 **Scientist-II**, *Institute for Infocomm Research (I2R), Agency for Science, Technology And Research (A*STAR)*, Singapore.

2019-2021 **Research Fellow**, *Satellite Research Centre (SaRC), Nanyang Technological University (NTU)*, Singapore.

2018-2019 **Research Fellow**, *Centre of Excellence for Testing and Research of AVs - NTU (CETRAN), ERIAN, Nanyang Technological University (NTU)*, Singapore.

2016-2018 **Research Fellow**, *ST Engineering - NTU Robotics Corporate Laboratory, Nanyang Technological University (NTU)*, Singapore.

Education

2011-2016 **Ph.D.**, *Indian Institute of Technology (IIT) Bombay*, Mumbai, India.

2008-2010 **M.Tech., Information Technology**, *International Institute of Information Technology (IIIT) Bangalore*, Bengaluru, India.

2006-2008 **M.Sc., Electronics**, *Andhra University*, Visakhapatnam, India.

2003-2006 **B.Sc., Electronics**, *Andhra University*, Visakhapatnam, India.

Awards & Honours

2017 Recipient of **the ST-NTU Conference Travel Grant** to attend ITSC-2017.

2015 Recipient of **the IIT Bombay Conference Travel Grant** to attend ICC-2015.

2008-2010 Recipient of **IIIT-B ABB Merit Scholarship** for M.Tech.

2008 Stood as the **University Topper** for M.Sc.(Electronics).

2006 Secured 26^{th} **All India Rank** in the university entrance test for M.Sc.(Electronics).

2004-2005 Recipient of **Best Student Award** for B.Sc.

2003 Recipient of **Pratibha Merit Scholarship** for Junior College.
2001 Recipient of **National Merit Scholarship** for Class-10.

Research Interests

- Multi-Robot Coordination and Mapping
- AI and optimization techniques
- Robotics and Autonomous Systems
- Intelligent Transport Systems
- Data Analytics and Simulations

Research Experience

2022-2026 **Institute for Infocomm Research (I2R), A*STAR,** Senior Scientist-I.
Topic: Trajectory-Based Operations (TBO) and Trajectory Prediction for Singapore Terminal Maneuvering Area (TMA).

- Developed a large-scale decision-support system that integrated real-time aircraft surveillance data and operational flight plans with simulation-based trajectory optimization.

- Designed KPI analysis framework for the TBO (Release 2) work, proposed a structured set of performance metrics, and carried out data analysis to compute and evaluate these KPIs.

- Built a hybrid ML model (decision trees + logistic regression) for air traffic controller workload computation, leading to a filed Singapore patent.

- Spearheaded the ATFM-ATM What-If Capability (WIC) project's UI prototype development efforts by supervising technical workflows and iterative design with an external UI/UX vendor.

- Built ML-based aircraft trajectory prediction models using aircraft historical trajectory data, intent, and weather to support arrival/departure flow management at Changi airport.

- Simulated airport surface movements for arrival and departure operations at Changi airport, proposed and validated several key performance metrics, and developed a Grafana-based dashboard integrated with a PostgreSQL database to visualize and track these KPIs.

- Led collaboration with key external stakeholders (CAAS, SIA, and MITRE) to align system capabilities with their operational workflows and validated prototypes via HITL exercises.

2019-2021 **Satellite Research Centre (SaRC), NTU,** Intra-CREATE Grant, Postdoctoral fellow.
Topic: Space Debris Removal from Low Earth Orbits (LEO) using CubeSats equipped with Robotic Arms and Vision.

- Performed a comprehensive literature survey on space debris tracking, rendezvous, capture, and deorbit mechanisms suitable for implementation using miniature satellites (CubeSats).

- Simulated deorbit trajectories to identify viable payload sizes achievable with CubeSats.

- Performed orbital decay analysis for the debris deorbiting CubeSat, accounting for the influence of key environmental parameters.

- Identified CubeSat-compatible debris capture and deorbit mechanisms from the literature.

- Built a CubeSat prototype equipped with a pair of robotic arms and vision sensors to capture and deorbit the target debris object (XSat) from LEO.

- Developed, trained, and deployed a YOLO-based real-time object detection model on Raspberry Pi to identify target debris object using on-board vision sensors.

- Engineered a dual 4-DoF robotic arm system using compact servo motors and vision-based tracking and actuation, optimized for integration into microsatellite platforms.

2018-2019 **Energy Research Institute @ NTU (ERIAN), NTU,** Postdoctoral fellow.
Topic: Designing Travel Time Prediction Algorithms for Autonomous Vehicles.

- Reviewed supervised and deep learning approaches for travel time prediction.
- Implemented several supervised learning methods for AV travel time prediction.
- Implemented LSTM-based travel time prediction algorithm for Dial-A-Ride Problem.

2016-2018 **ST Engineering - NTU Robotics Corporate Laboratory,** Postdoctoral fellow.
Topic: Optimization, Scheduling, Allocation and Routing for Autonomous Vehicles.

- Designed new meta-heuristic optimization algorithms for vehicle routing by leveraging Tabu Search, Ant Colony Optimization, Genetic Algorithms, and Simulated Annealing.
- Implemented travel time and demand prediction models using supervised machine learning.
- Designed a modular software architecture and built a high-fidelity Vissim-based simulation engine to test vehicle routing algorithms at large scale.
- Developed algorithms for fleet size minimization, disrupted ride sharing problem, finding optimal depot locations for autonomous vehicles in large scale fleet management systems.

2011-2016 **Indian Institute of Technology (IIT) Bombay,** PhD research scholar.
Topic: Multi-Robot Exploration and Map Building with Assured Collision Avoidance.

- Developed a new decentralized multi-robot exploration and mapping algorithm, *Improved Multi-Robot Depth First Search (IMRDFS)* and validated it using simulations.
- Proposed a new data structure, *modified incidence matrix (MIM)* to store and update the map of the environment, to achieve coordinated and non-redundant exploration.
- Created a generic framework for multi-agent graph exploration and mapping.
- Developed a decentralized collision avoidance algorithm for multi-robot exploration.
- Experimentally validated these algorithms using a team of mobile robots.

2009-2010 **ABB Indian Corporate Research Center, Bangalore,** Master's research project.

- Performed software reliability modelling for ABB Robot Studio using software failure data.
- Arm manipulation and motion planning for the ABB IRB-1600 robotic arm for pick and place of objects, and painting tasks.

Communication & Leadership

- Led multi-stakeholder Human In The Loop (HITL) validation exercise with 20+ participants.
- Supervised UI/UX design and stakeholder engagement for multiple projects at I2R, A*STAR.
- Mentored and guided one Ph.D. candidate along with multiple master's, undergraduate, and internship students on applied AI and optimization research projects.

Technical Proficiency

- Core Skills : AI/ML, optimization, Predictive Modeling, Simulation, Data Analytics.
- Programming/Scripting : Matlab, Python, C++.
- Tools/Software : Tableau, Grafana, Postgres DB, Vissim, AirTOP, ROS, LaTeX.
- ML/Python Libraries : Scikit-learn, PyTorch, Keras, Pandas, NumPy, Seaborn, Matplotlib.

Patents

- Tanaya Chaudhuri, Zhang Sheng, Zhang Yicheng, **Sarat Chandra Nagavarapu** and Jaya Shankar, "*Apparatus and Method for Predicting Air Traffic Controller Workload,*" (patent filed, pending grant), Agency for Science, Technology and Research, Singapore.

Publications

Journals

- **Sarat Chandra Nagavarapu**, Leena Vachhani and Arpita Sinha, "Multi-Robot Graph Exploration and Map Building with Collision Avoidance: A Decentralized Approach," *Journal of Intelligent and Robotic Systems*, vol. 83, no. 3, pp. 503-523, Springer, 2016.

- **Sarat Chandra Nagavarapu**, Leena Vachhani, Arpita Sinha and Somnath Buruily, “Generalizing Multi-Agent Graph Exploration Techniques,” *International Journal of Control, Automation and Systems*, vol. 19, no. 1, pp. 491-504, Springer, 2021.
- **Sarat Chandra Nagavarapu**, Laveneishyan B. Mogan, Amal Chandran and Daniel E. Hastings, “CubeSats for Space Debris Removal from LEO: Prototype Design of a Robotic Arm-based Deorbiter CubeSat,”, *Advances in Space Research*, vol. 75, no. 9, pp. 6896-6910, Elsevier, 2025.
- Ramesh Ramasamy Pandi, Song Guang Ho, **Sarat Chandra Nagavarapu** and Justin Dauwels, “A Generic GPU-Accelerated Framework for the Dial-A-Ride Problem,” *IEEE Transactions on Intelligent Transportation Systems*, vol. 22, no. 10, pp. 6473-6488, 2021.
- Ramesh Ramasamy Pandi, Song Guang Ho, **Sarat Chandra Nagavarapu**, Twinkle Tripathy and Justin Dauwels, “Disruption Management for Dial-A-Ride Systems,” *IEEE Intelligent Transportation Systems Magazine*, vol. 12, no. 4, pp. 219-234, 2020.
- Song Guang Ho, **Sarat Chandra Nagavarapu**, Ramesh Ramasamy Pandi, Kaveh Azizian and Justin Dauwels, “An Improved Tabu Search Heuristic for Static Dial-A-Ride Problem,” (arXiv preprint available).

Book Chapters

- **Sarat Chandra Nagavarapu**, Anuj Abraham, Sihao Li and Justin Dauwels, “Enhancing Autonomous Vehicle Navigation Through Computer Vision: Techniques for Lane Marker Detection and Rain Removal,” *Internet of Vehicles and Computer Vision Solutions for Smart City Transformations. Lecture Notes in Intelligent Transportation and Infrastructure* (LNITI), pp. 167-191, Springer, Cham, 2025.
- Twinkle Tripathy, **Sarat Chandra Nagavarapu**, Kaveh Azizian, Ramesh Ramasamy Pandi and Justin Dauwels, “Solving Dial-A-Ride Problems using Multiple Ant Colony System with Fleet Size Minimisation,” *Advances in Computational Intelligence Systems* (ACIS), vol. 650, pp. 325-336, Springer, 2018.
- Mohan Kashyap Pargi, Anuj Abraham and **Sarat Chandra Nagavarapu**, “Transforming Public Safety and QoL in Smart Cities: Automated Event Detection and Response Generation with AI and IoT,” *Internet of Vehicles and Computer Vision Solutions for Smart City Transformations. Lecture Notes in Intelligent Transportation and Infrastructure* (LNITI), pp. 267-290, Springer, Cham, 2025.

Conferences

- **Sarat Chandra Nagavarapu**, Leena Vachhani and Arpita Sinha, “A Decentralized Approach for Autonomous Multi-Robot Exploration and Map Building for Tree Structures,” in proceedings of the *Indian Control Conference (ICC)*, Chennai, India, pp. 274-279, 2015.
- **Sarat Chandra Nagavarapu**, Twinkle Tripathy and Justin Dauwels, “Development of a Simulation Platform to Implement Vehicle Routing Algorithms for Large Scale Fleet Management Systems,” in proceedings of the 20^{th} *IEEE International Conference on Intelligent Transportation Systems (ITSC): Workshop*, Yokohama, Japan, pp. 124-129, 2017.
- Song Guang Ho, **Sarat Chandra Nagavarapu**, Ramesh Ramasamy Pandi and Justin Dauwels, “Improved Tabu Search Heuristic for Static Dial-a-ride Problem: Faster and Better Convergence,” presented at the 25^{th} *Intelligent Transportation Systems World Congress (ITS-WC)*, Copenhagen, Denmark, 2018.
- Song Guang Ho, Ramesh Ramasamy Pandi, **Sarat Chandra Nagavarapu** and Justin Dauwels, “Multi-atomic Annealing Heuristic for the Dial-a-ride Problem,” in proceedings of the 12^{th} *IEEE International Conference on Service Operations and Logistics, and Informatics (SOLI)*, Singapore, pp. 272-277, 2018.
- Ramesh Ramasamy Pandi, Song Guang Ho, **Sarat Chandra Nagavarapu**, Twinkle Tripathy and Justin Dauwels, “GPU-Accelerated Tabu Search Algorithm for Dial-A-Ride Problem,” in proceedings of the 21^{st} *IEEE International Conference on Intelligent Transportation Systems (ITSC)*, Hawaii, USA, pp. 2519-2524, 2018.
- Song Guang Ho, Hong Wei Koh, Ramesh Ramasamy Pandi, **Sarat Chandra Nagavarapu** and Justin Dauwels, “Solving Time-Dependent Dial-A-Ride Problem using Greedy Ant Colony Optimization,” in proceedings of the 21^{st} *IEEE International Conference on Intelligent Transportation Systems (ITSC)*, Hawaii, USA, pp. 778-785, 2018.

- Ramesh Ramasamy Pandi, Song Guang Ho, **Sarat Chandra Nagavarapu** and Justin Dauwels, "Deterministic Annealing for Depot Optimization: Applications to the Dial-A-Ride Problem," in proceedings of the 9^{th} *IEEE Symposium Series on Computational Intelligence (SSCI)*, Bangalore, India, pp. 88-95, 2018.
- Ramesh Ramasamy Pandi, Song Guang Ho, **Sarat Chandra Nagavarapu** and Justin Dauwels, "Adaptive Algorithm for Dial-A-Ride Problem with Vehicle Breakdown," in proceedings of the 18^{th} *European Control Conference (ECC)*, Naples, Italy, pp. 2682-2688, 2019.
- Aakash Kumar Narayanan, Chaitra Pranesh, **Sarat Chandra Nagavarapu**, B. Anil Kumar and Justin Dauwels, "Data-driven Models for Short-term Travel Time Prediction," in proceedings of the 22^{nd} *IEEE International Conference on Intelligent Transportation Systems (ITSC)*, Auckland, New Zealand, pp. 1941-1946, 2019.
- **Sarat Chandra Nagavarapu**, Amal Chandran and Daniel E. Hastings, "Orbital Decay Analysis for Debris Deorbiting CubeSats in LEO: A Case Study for the VELOX-II Deorbit Mission," in proceedings of the 8^{th} *European Conference on Space Debris*, Darmstadt, Germany, 2021.
- Libin K. Mathew, **Sarat Chandra Nagavarapu** and Anuj Abraham, "Design and implementation of a Joint Sensing & Communication System for Connected Autonomous Vehicles," in proceedings of the 17^{th} *International Conference on Control, Automation, Robotics and Vision (ICARCV)*, Singapore, pp. 392-396, 2022.
- Anuj Abraham, **Sarat Chandra Nagavarapu**, Shitala Prasad, Pranjal Vyas and Libin K. Mathew, "Recent Trends in Autonomous Vehicle Validation Ensuring Road Safety with Emphasis on Learning Algorithms," in proceedings of the 17^{th} *International Conference on Control, Automation, Robotics and Vision (ICARCV)*, Singapore, pp. 397-404, 2022.
- **Sarat Chandra Nagavarapu**, Laveneishyan B. Mogan, Amal Chandran and Daniel E. Hastings, "A Robotic Arm-Based Deorbiter CubeSat for Space Debris Removal from LEO," presented at the 5^{th} *COSPAR Symposium on Space Science with Small Satellites*, Singapore, 16-21 April, 2023.
- **Sarat Chandra Nagavarapu**, Anuj Abraham, Nithish Selvaraj Muthuchamy and Justin Dauwels, "A Dynamic Object Removal and Reconstruction Algorithm for Point Clouds,", in proceedings of the 17^{th} *International Conference on Service Operations and Logistics, and Informatics (SOLI)*, Singapore, pp. 1-8, 2023.
- **Sarat Chandra Nagavarapu**, Leena Vachhani and Arpita Sinha, "A Fail-Safe Graph Exploration Algorithm for Large-Scale Multi-Robot Systems," (in preparation).
- **Sarat Chandra Nagavarapu**, Leena Vachhani and Arpita Sinha, "A Beam Steering based Framework for Continuous Target Tracking in Multi-Robot Systems," (in preparation).

Invited Talks

- ***Fleet Management Systems with Special Emphasis on Vehicle Routing Algorithms***, ST Engineering - NTU Robotics Corporate Laboratory, Nanyang Technological University, Singapore, 30 August, 2018.
- ***Multi-Robot Graph Exploration and Mapping with Collision Avoidance***, Department of Electronics, Govt. Degree College (Men), Srikakulam, 4 August, 2018.
- ***What is expected out of a Ph.D.: Synopsis/Thesis Layout***, Communication Skills II (SC-792), Systems & Control Engineering, IIT Bombay, 9 March, 2016.
- ***Introduction to Autonomous Robotic Systems***, Students' Technical Activities Body (STAB), IIT Bombay, 28 January, 2014.
- ***Hands-on Tutorial for Coordination Algorithms using Player-Stage Software***, QIP short term course on Motion Planning and Coordination of Autonomous Robots, IIT Bombay, 7-11 May, 2012.

Professional Memberships

- *Member*, IEEE
- IEEE Robotics and Automation Society, IEEE Multi-Robot Systems.

Synergistic Activities

- Served as deputy scientific officer (DSO) and Technical Programme Committee (TPC) member for multiple conferences, serving as a reviewer for multiple journals and conferences.